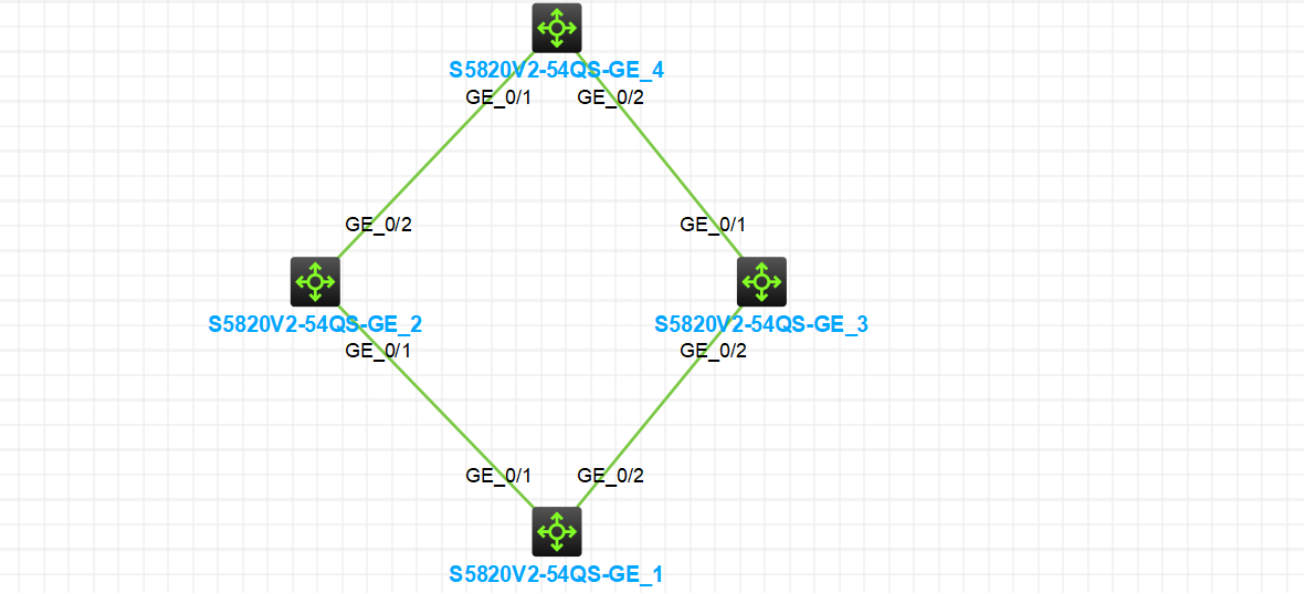


Smart-Link实验

实验拓扑

- 实验需求
- 1.网络中有 Vlan10，Vlan20，Vlan30，Vlan40 一共 4 个业务 Vlan，数据可以在所有链路上转发
 - 2.在 SW1 上使用 Smart-link 对上行链路进行备份。要求 Smart-link组1 引用 MST 实例 1（映射 Vlan10 和 Vlan20）的流量经过 SW2 的链路访问上行设备，而 Smart-link 组 2 引用实例 2（映射 Vlan30 和 Vlan40）的流量经过 SW3 的链路访问上行设备，组 1 和组 2 分别在 Vlan100 和 Vlan200 内发送和接收 Flush 报文
 - 3.要求当 Smart-link 主链路故障恢复后，端口角色能够自动恢复
 - 4.要求当 SW2 和 SW3 的上行链路故障后，也能触发 Smart-link 角色切换



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1. 网络中有 Vlan10，Vlan20，Vlan30，Vlan40 一共 4 个业务 Vlan，数据可以在所有链路上转发
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3. 要求当 Smart-link 主链路故障恢复后，端口角色能够自动恢复
4. 要求当 SW2 和 SW3 的上行链路故障后，也能触发 Smart-link 角色切换

实验配置

```
1 [SW1]vlan 10
2 [SW1-vlan10]vlan 20
3 [SW1-vlan20]vlan 30
4 [SW1-vlan30]vlan 40
5 [SW1-vlan40]vlan 100
```

```
6 [SW1-vlan100]vlan 200
7 [SW1-vlan200]int g1/0/1
8 [SW1-GigabitEthernet1/0/1]port link-type trunk
9 [SW1-GigabitEthernet1/0/1]port trunk permit vlan 10 20 100
10 [SW1-GigabitEthernet1/0/1]int g1/0/2
11 [SW1-GigabitEthernet1/0/2]port link-type trunk
12 [SW1-GigabitEthernet1/0/2]port trunk permit vlan 30 40 200
13 [SW1-GigabitEthernet1/0/2]int ra g 1/0/1 to g 1/0/2
14 [SW1-if-range]un stp enable
15 ---
16 [SW2]vlan 10
17 [SW2-vlan10]vlan 20
18 [SW2-vlan20]vlan 100
19 [SW2-vlan100]int g1/0/1
20 [SW2-GigabitEthernet1/0/1]port link-type trunk
21 [SW2-GigabitEthernet1/0/1]port trunk permit vlan 10 20 100
22 [SW2-GigabitEthernet1/0/1]int g1/0/2
23 [SW2-GigabitEthernet1/0/2]port link-type trunk
24 [SW2-GigabitEthernet1/0/2]port trunk permit vlan 10 20 100
25 [SW2-GigabitEthernet1/0/2]
26 [SW2-GigabitEthernet1/0/2]
27 ---
28 [SW3]vlan 30
29 [SW3-vlan30]vlan 40
30 [SW3-vlan40]vlan 200
31 [SW3-vlan200]int g1/0/1
32 [SW3-GigabitEthernet1/0/1]port link-type trunk
33 [SW3-GigabitEthernet1/0/1]port trunk permit vlan 30 40 200
34 [SW3-GigabitEthernet1/0/1]int g1/0/2
35 [SW3-GigabitEthernet1/0/2]port link-type trunk
36 [SW3-GigabitEthernet1/0/2]port trunk permit vlan 30 40 200
37 [SW3-GigabitEthernet1/0/2]
38 [SW3-GigabitEthernet1/0/2]
39 ---
40 [SW4]vlan 10
41 [SW4-vlan10]vlan 20
42 [SW4-vlan20]vlan 30
43 [SW4-vlan30]vlan 40
44 [SW4-vlan40]vlan 100
45 [SW4-vlan100]vlan 200
46 [SW4-vlan200]int g1/0/1
47 [SW4-GigabitEthernet1/0/1]port link-type trunk
48 [SW4-GigabitEthernet1/0/1]port trunk permit vlan 10 20 100
49 [SW4-GigabitEthernet1/0/1]int g1/0/2
50 [SW4-GigabitEthernet1/0/2]port link-type trunk
51 [SW4-GigabitEthernet1/0/2]port trunk permit vlan 30 40 200
52 [SW4-GigabitEthernet1/0/2]
53 [SW4-GigabitEthernet1/0/2]
54
55
56 [SW1]stp region-configuration 1
57 [SW1-mst-region]region-name 1
58 [SW1-mst-region]revision-level 1
```

```

59 [SW1-mst-region]instance 1 vlan 10 20 100
60 [SW1-mst-region]instance 2 vlan 30 40 200
61 [SW1-mst-region]active region-configuration
62 [SW1]smart-link group 1
63 [SW1-smlk-group1]protected-vlan reference-instance 1
64 [SW1-smlk-group1]flush enable control-vlan 100
65 [SW1-smlk-group1]port g1/0/1 primary
66 [SW1-smlk-group1]port g1/0/2 secondary
67 [SW1]smart-link group 2
68 [SW1-smlk-group2]protected-vlan reference-instance 2
69 [SW1-smlk-group2]flush enable control-vlan 200
70 [SW1-smlk-group2]port g1/0/2 primary
71 [SW1-smlk-group2]port g1/0/1 secondary
72 [SW1]smart-link group 1
73 [SW1-smlk-group1]protected-vlan reference-instance 0
74 [SW1]smart-link group 1
75 [SW1-smlk-group1]preemption mode role
76 [SW1]smart-link group 2
77 [SW1-smlk-group2]preemption mode role
78 ---
79 [SW2]int range g 1/0/1 to g 1/0/2
80 [SW2-if-range]smart-link flush enable control-vlan 100 200
81 [SW2]monitor-link group 1
82 [SW2-mtlk-group1]port g1/0/2 uplink
83 [SW2-mtlk-group1]port g1/0/1 downlink
84 ---
85 [SW3]int range g 1/0/1 to g 1/0/2
86 [SW3-if-range]smart-link flush enable control-vlan 100 200
87 [SW3]monitor-link group 1
88 [SW3-mtlk-group1]port g1/0/1 uplink
89 [SW3-mtlk-group1]port g1/0/2 downlink
90 ---
91 [SW4]int range g 1/0/1 to g 1/0/2
92 [SW4-if-range]smart-link flush enable control-vlan 100 200
93
94

```

结果验证

1.查看smart-link信息

```

1 <SW1>display smart-link group all
2 Smart link group 1 information:
3   Device ID       : 48ff-3264-0100
4   Preemption mode : Role
5   Preemption delay: 1(s)
6   Control VLAN    : 100
7   Protected VLAN  : Reference Instance 0 to 1
8

```

9	Member	Role	State	Flush-count	Last-flush-time
10	-----				
11	GE1/0/1	PRIMARY	ACTIVE	5	00:10:22 2025/11/13
12	GE1/0/2	SECONDARY	STANDBY	0	NA
13					
14	Smart link group 2 information:				
15	Device ID : 48ff-3264-0100				
16	Preemption mode : Role				
17	Preemption delay: 1(s)				
18	Control VLAN : 200				
19	Protected VLAN : Reference Instance 2				
20					
21	Member	Role	State	Flush-count	Last-flush-time
22	-----				
23	GE1/0/2	PRIMARY	ACTIVE	6	00:10:22 2025/11/13
24	GE1/0/1	SECONDARY	STANDBY	0	NA

2.将SW2的1/0/2端口shutdown再查看Smart-Link

1	<SW1>display smart-link group all				
2	Smart link group 1 information:				
3	Device ID : 48ff-3264-0100				
4	Preemption mode : Role				
5	Preemption delay: 1(s)				
6	Control VLAN : 100				
7	Protected VLAN : Reference Instance 0 to 1				
8					
9	Member	Role	State	Flush-count	Last-flush-time
10	-----				
11	GE1/0/1	PRIMARY	DOWN	5	00:10:22 2025/11/13
12	GE1/0/2	SECONDARY	ACTIVE	0	NA
13					
14	Smart link group 2 information:				
15	Device ID : 48ff-3264-0100				
16	Preemption mode : Role				
17	Preemption delay: 1(s)				
18	Control VLAN : 200				
19	Protected VLAN : Reference Instance 2				
20					
21	Member	Role	State	Flush-count	Last-flush-time
22	-----				
23	GE1/0/2	PRIMARY	ACTIVE	6	00:10:22 2025/11/13
24	GE1/0/1	SECONDARY	DOWN	0	NA

3.将SW3的1/0/1端口也shutdown再查看Smart-Link

1	<SW1>display smart-link group all				
2	Smart link group 1 information:				
3	Device ID : 48ff-3264-0100				
4	Preemption mode : Role				
5	Preemption delay: 1(s)				

```
6 Control VLAN : 100
7 Protected VLAN : Reference Instance 0 to 1
8
9 Member Role State Flush-count Last-flush-time
10 -----
11 GE1/0/1 PRIMARY DOWN 5 00:10:22 2025/11/13
12 GE1/0/2 SECONDARY DOWN 0 NA
13
14 Smart link group 2 information:
15 Device ID : 48ff-3264-0100
16 Preemption mode : Role
17 Preemption delay: 1(s)
18 Control VLAN : 200
19 Protected VLAN : Reference Instance 2
20
21 Member Role State Flush-count Last-flush-time
22 -----
23 GE1/0/2 PRIMARY DOWN 6 00:10:22 2025/11/13
24 GE1/0/1 SECONDARY DOWN 0 NA
```

4.将shudown的端口un shudown后，再把SW4的两个端口都shudown

```
1 <SW1>display smart-link group all
2 Smart link group 1 information:
3 Device ID : 48ff-3264-0100
4 Preemption mode : Role
5 Preemption delay: 1(s)
6 Control VLAN : 100
7 Protected VLAN : Reference Instance 0 to 1
8
9 Member Role State Flush-count Last-flush-time
10 -----
11 GE1/0/1 PRIMARY ACTIVE 7 00:32:39 2025/11/13
12 GE1/0/2 SECONDARY STANDBY 0 NA
13
14 Smart link group 2 information:
15 Device ID : 48ff-3264-0100
16 Preemption mode : Role
17 Preemption delay: 1(s)
18 Control VLAN : 200
19 Protected VLAN : Reference Instance 2
20
21 Member Role State Flush-count Last-flush-time
22 -----
23 GE1/0/2 PRIMARY ACTIVE 7 00:32:43 2025/11/13
24 GE1/0/1 SECONDARY STANDBY 0 NA
25 ---
26 <SW1>display smart-link group all
27 Smart link group 1 information:
28 Device ID : 48ff-3264-0100
29 Preemption mode : Role
30 Preemption delay: 1(s)
```

31	Control VLAN	:	100		
32	Protected VLAN	:	Reference Instance 0 to 1		
33					
34	Member	Role	State	Flush-count	Last-flush-time
35	-----				
36	GE1/0/1	PRIMARY	DOWN	7	00:32:39 2025/11/13
37	GE1/0/2	SECONDARY	DOWN	0	NA
38					
39	Smart link group 2 information:				
40	Device ID	:	48ff-3264-0100		
41	Preemption mode	:	Role		
42	Preemption delay	:	1(s)		
43	Control VLAN	:	200		
44	Protected VLAN	:	Reference Instance 2		
45					
46	Member	Role	State	Flush-count	Last-flush-time
47	-----				
48	GE1/0/2	PRIMARY	DOWN	7	00:32:43 2025/11/13
49	GE1/0/1	SECONDARY	DOWN	0	NA

附录

Smart-link 配置完成后，由于只引用了实例1和实例2，并没有引用Vlan1所在的实例0，实例0由MSTP的IST来进行防环；而SW1的相关接口关闭了STP特性，导致SW2、SW3和SW4上的STP协议判断没有环路，从而不会阻塞任何端口；而且所有接口都配置为了Trunk并默认放行Vlan1，所以当前网络中Vlan1存在逻辑环路；这里采取把Vlan1所在的实例0也引用到Smart-link组1来解决